9.6 Permutations and Combinations

Name _____ Period ____ Date _____

Algebra I 9.6 · Student sheet

I can…

- count the ways to arrange objects using factorials.
- find permutations and combinations and tell which one to use.
- use permutations and combinations to find probabilities.

Vocabulary · fill in the missing word

n factorial (n!) — $n \cdot (n-1) \cdot \ldots \cdot 2 \cdot 1$; $0! =$ _____

Permutation — an _____ in which order matters

$_nP_r$ — $n!/(n-r)!$, the number of _____ of r objects chosen from n

Combination — a _____ in which order does not matter

$_nC_r$ — $n!/((n-r)! \cdot r!)$, the number of _____ of r objects chosen from n

Part 1 · Officers vs. committee

Your group is A, B, C, and D.

1. List every way to choose a **president and a vice president** (write president first).

2. List every way to choose a **2-person committee**.

How many in each list? _____ and _____. Why is one list twice as long?

Part 2 · Notes and examples (fill in each note, then work the example)

Factorial: n! = _____ 0! = _____

Example 1

a. In how many ways can you arrange the letters in MATH?

b. In how many ways can you arrange 2 of the letters?

Permutations: Order matters → _____: $_nP_r =$ _____

Example 2

Ten runners are in a race. In how many ways can the gold, silver, and bronze medals be awarded?

Combinations: Order doesn't matter → _____: $_nC_r =$ _____

Example 3

A pizza shop has 8 toppings. How many different 3-topping pizzas can you order?

Example 4

From a club of 12 members, how many ways can you

a. choose a 4-member committee?

b. choose a captain and a co-captain?

Probability with counting: Probability = (favorable selections) ÷ _____

Example 5

A class has 5 boys and 7 girls. Three students are chosen at random to present. What is the probability that all three are girls?

Part 3 · Partner practice

Partner 1

a. In how many ways can 6 books be arranged on a shelf?

b. In how many ways can 3 of the 6 books be arranged?

Partner 2

A pizza shop offers 10 toppings. How many different 3-topping pizzas are possible?